The energy that becomes thermal

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for mechanical energy changes through heating and deformation.

SUCCESS CRITERIA - I can...

1. Track system boundaries in an energy balance.
2. Distinguish thermal energy from temperature.
3. Use energy differences to quantify a transfer.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “thermal energy” and “system” to explain your answer. Sentence starters are optional.

thermal energy: internal energy associated with microscopic motion/interactions

system: the object or objects chosen for analysis

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

A spring stores **3 J** at a given compression. Within its Hooke’s-law range, find the energy at **5** times that compression.

Word help: compression = shortening or squeezing relative to an unstressed length

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $E_{mechanical,\ i} = E_{mechanical,\ f} + \Delta E_{thermal}$.
No other transfers in this calculation; include contacting materials in the system. Friction transfers mechanical energy to internal energy.

Worked reasoning: A **2 N** friction force opposes a sliding box over **5 m**. It does **-10 J** of work; in a suitable box-floor system this appears as a **10 J** thermal-energy increase.

Your turn: 1. **Draw the force**, displacement and energy transfer.
2. **Name the system** for your energy account.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial **Transfers** **Final**

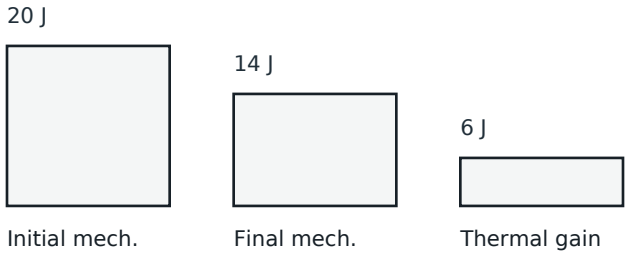
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A cart begins with **20 J** mechanical energy and ends with **14 J**. The closed accounting includes **6 J** thermal increase. Does it balance?

Sentence starter: I notice … . This supports … because … .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A crate slides **3 m** while a constant **5 N** friction force opposes its motion.
1. **Draw the force** and displacement.
2. **Find friction's work** and the thermal energy increase of the crate-floor system if there are no other energy changes.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Does the same energy input always produce the same temperature rise in two different objects? **Name quantities you would need**.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart's work is measured over the first **4 m**, but its speed change is measured over **6 m**. Can these values directly test $W_{net}=\Delta K$? Explain.
Show the energy changes and transfers; **label units**.

Word help: work = energy transferred by a force along a displacement

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

An energy account has **12 J** initially, **7 J** motion energy and **3 J** thermal energy finally. What is unaccounted for? **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A box starts with **20 J** of kinetic energy and stops under **4 N** friction.
1. **Find stopping distance** and thermal-energy increase.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3